

PCS-302 : Data Structures Lab

Reverse a Queue | Using a Stack

Problem Statement

Design an algorithm and write a program to reverse a queue.

Example

Original Queue (front to rear): 10 20 30 40

Step 1 (dequeue -> push): push 10, push 20, push 30, push 40
Stack (top to bottom): 40 30 20 10

Step 2 (pop -> enqueue): enqueue 40, 30, 20, 10

Reversed Queue (front to rear): 40 30 20 10

Algorithm (Using a Stack)

```
ReverseQueue():
    1. While queue is not empty:
        dequeue an element from the queue
        push it onto a stack

    2. While stack is not empty:
        pop an element from the stack
        enqueue it back into the queue
```

A stack is LIFO, so moving every element queue -> stack -> queue naturally flips the order.

C Program

```
#include <stdio.h>
#define MAX 10

int queue[MAX];
int front = 0, rear = -1, count = 0;

int stack[MAX];
int top = -1;

void enqueue(int k) {
    if (rear == MAX - 1) {
        printf("Queue Overflow.\n");
        return;
    }
    rear++;
    queue[rear] = k;
    count++;
}

int dequeue() {
```

```

        int val = queue[front];
        front++;
        count--;
        return val;
    }

    int isEmpty() {
        return (count == 0);
    }

    void push(int k) {
        top++;
        stack[top] = k;
    }

    int pop() {
        int val = stack[top];
        top--;
        return val;
    }

    void printQueue() {
        int i;
        printf("Queue (front to rear): ");
        for (i = front; i <= rear; i++)
            printf("%d ", queue[i]);
        printf("\n");
    }

    void reverseQueue() {
        // Step 1: move every element from queue to stack
        while (isEmpty() == 0) {
            int val = dequeue();
            push(val);
        }

        // Step 2: move every element back from stack to queue
        while (top != -1) {
            int val = pop();
            enqueue(val);
        }
    }

    int main() {
        enqueue(10);
        enqueue(20);
        enqueue(30);
        enqueue(40);

        printf("Original ");
        printQueue();

        reverseQueue();

        printf("Reversed ");
    }

```

```
    printQueue();  
  
    return 0;  
}
```